Homework 3

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1. $\frac{d}{dx}\left(\frac{1}{1+e^x} - \frac{1}{2}\right) = -(1+e^x)^{-2} \cdot e^x = \boxed{\frac{-e^x}{(1+e^x)^2}}$

I used chain rule to get this. It states that $d/dx(f(g(x))) = f'(g(x)) \cdot g'(x)$ so $d/dx((1+e^x)^{-1}) = -(1+e^x)^{-2} \cdot e^x$ where $f(x) = x^{-1}$ and $g(x) = 1+e^x$. The other term, $d/dx(-1/2)$, can be ignored because the derivative of a constant is 0.

4. The largest interval for which all initial guesses converge in Newton's method for this function is $[-2.177368, 2.177368]$.

[52]:

```python
import numpy as np
import math
def f(x):
    return (1/(1+math.exp(x)))-(1/2)
def fprime(x):
    ans=0
    try:
        ans = (-math.exp(x))/((1 + math.exp(x)) ** 2)
    except OverflowError:
        ans = None
    return ans
def newtonsMethod(x):
    h = f(x) / fprime(x)
    count = 0
    while abs(h) >= 0.0001:
        if fprime(x) == None:
            return None
        if fprime(x) == 0:
            return None
        h = f(x)/fprime(x)
        x = x - h
        count = count + 1
        if count > 1000:
            return None
    return x
```

[56]:

```python
def findInterval(x, increment):
    if increment < 0.0001:
        return x
    if newtonsMethod(x) == None:
        return findInterval(x-increment,increment/2)
    else:
        return findInterval(x+increment,increment/2)
#For this to work, we need an initial guess for x where either x converges and x + increment diverges,
#or where x diverges and x - increment converges.
#I tested x = 2 and x = 2.5 and 2 converged and 2.5 diverged so I used x = 2 as my initial guess
a = findInterval(2.0,0.5)
print("The interval in which Newton's method converges for this function is [%f,%f]"%(-a,a))
```

```
The interval in which Newton's method converges for this function is [-2.177368,2.177368]
```